



CELEBAL

TECHNOLOGIES

Celebal Summer Internship 2026

ASSIGNMENT – 2

SQL – BASED DATA ANALYSIS USING FILTERING, AGGREGATION, AND BUSINESS QUERIES

This project uses SQL to analyze business data by applying filtering, aggregation, and grouping techniques. The goal is to extract meaningful insights, identify trends, and support data-driven business decisions.

QUESTION 1:-

LOAD DATASET INTO SOL DATABASE

OBJECTIVE

To import the Superstore Sales dataset into MySQL and verify that the data has been loaded successfully.

PROCEDURE

1. Open MySQL Workbench

To establish a connection with the MySQL server and access the SQL environment.

Procedure

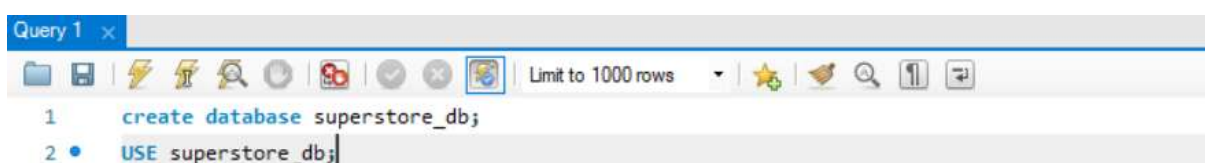
- Launch MySQL Workbench.
- Connect to the local MySQL server using the root account.
- Open a new SQL Editor tab.

Observation

A successful connection indicates that the MySQL server is running properly and ready for database operations.

2. Create a New Database

To create a dedicated database for storing the Superstore Sales dataset.



Explanation

CREATE DATABASE

Creates a new database named superstore_db.

USE

Selects the database so that all subsequent operations are performed within it.

OUTPUT :-



#	Time	Action	Message	Duration / Status
1	12:29:30	SHOW SESSION VARIABLES LIKE 'lower_case_table_names'	OK	0.000 sec
2	12:29:30	SHOW DATABASES	OK	0.000 sec
3	12:29:40	SHOW SESSION VARIABLES LIKE 'lower_case_table_names'	OK	0.000 sec
4	12:29:40	SHOW TABLES FROM 'superstore_db' like 'sample - superstore'	OK	0.000 sec
5	12:30:16	CREATE TABLE 'superstore_db'.'sample - superstore' ('Row ID' int, 'Order ID' int, 'Order Date' text, 'Ship Date' text ...	OK	0.000 sec
6	12:30:16	PREPARE stmt FROM INSERT INTO 'superstore_db'.'sample - superstore' ('Row ID' , 'Order ID' , 'Order Date' , 'Ship ...	OK	0.000 sec
7	12:32:15	DEALLOCATE PREPARE stmt	OK	0.000 sec

Observation

The database was created successfully and is ready to store the dataset.

3. Import the Dataset

- Right-click on Tables under the superstore_db schema.
- Select Table Data Import Wizard.
- Browse and select the superstore.csv file.
- Choose Create New Table and name it superstore_sales.
- Complete the import process.

OUTPUT:-

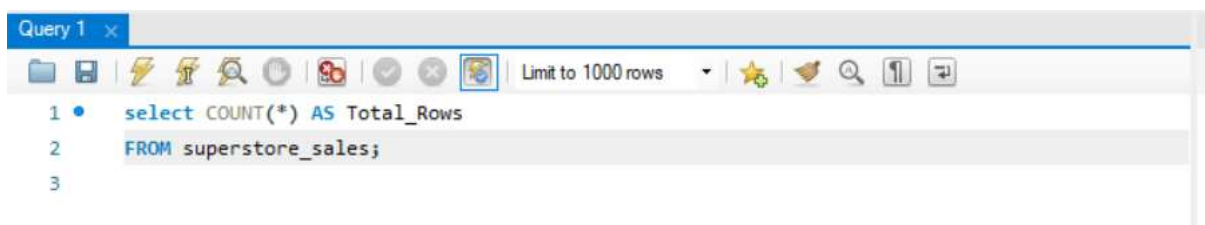


IMPORT COMPLETED SUCCESSFULLY

4. Verify Data Import

To confirm that all records have been imported correctly.

QUERY 1:- Count Total Records

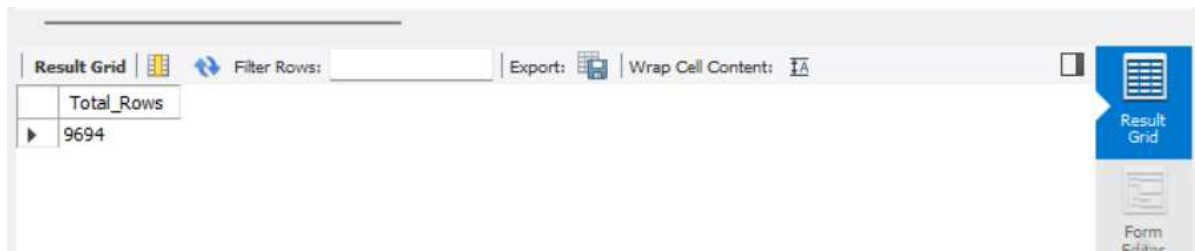


Explanation

COUNT(*)

Counts the total number of records present in the table.

OUTPUT:-



Observation

The table contains 9,694 records, confirming that the dataset was loaded successfully.

QUERY 2:- View Sample Records



Explanation

LIMIT 10

Displays the first 10 records from the dataset.

OUTPUT:-

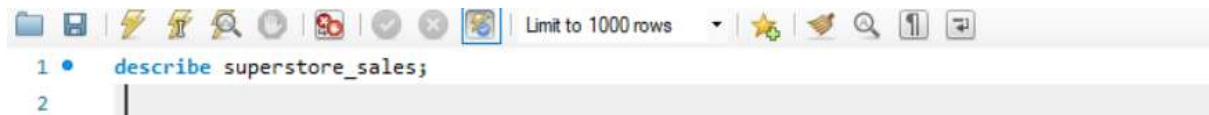
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer

Observation

The data is displayed correctly, indicating that the import process was successful.

5. Check Table Structure

To review the table schema and verify column names and data types.



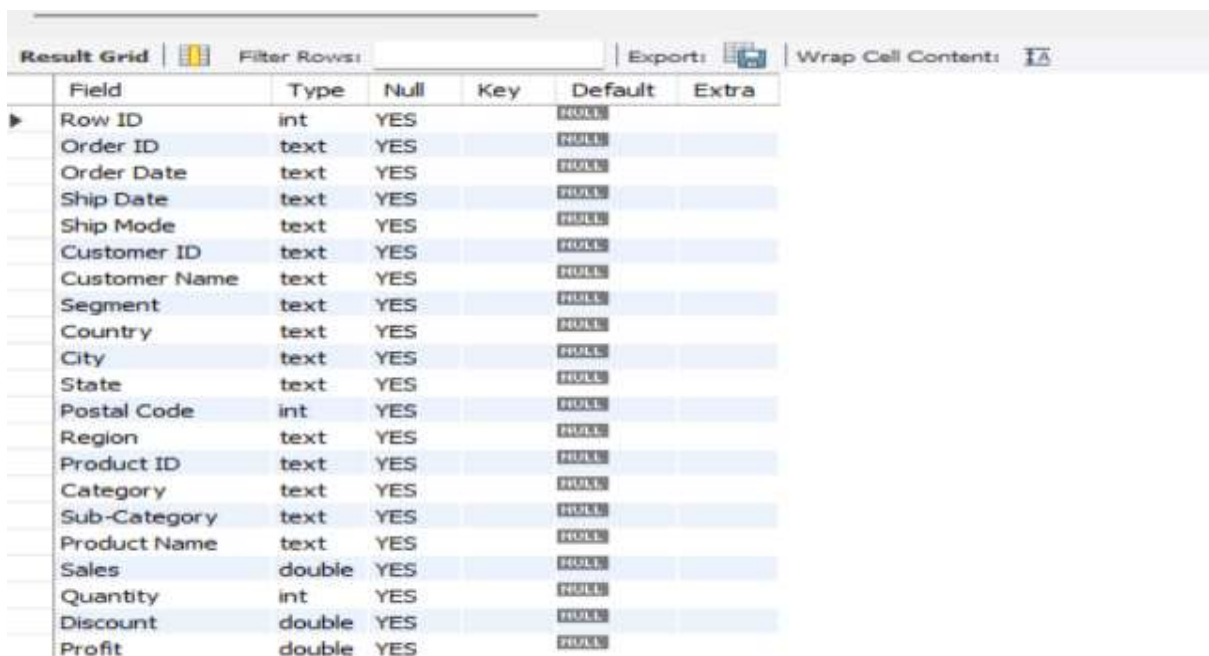
```
1 • describe superstore_sales;
2 |
```

Explanation

DESCRIBE

Displays all columns, data types, null constraints, and key information for the table.

OUTPUT:-



	Field	Type	Null	Key	Default	Extra
►	Row ID	int	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	text	YES		NULL	
	Ship Date	text	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	int	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

Observation

The table structure confirms that the required columns have been imported successfully.

Result

The Superstore Sales dataset was successfully imported into MySQL and verified using row count, sample records, and schema inspection.

Conclusion

The database and table were created successfully, and the dataset was imported without errors. Verification queries confirmed that the data was loaded correctly and is ready for exploration, filtering, aggregation, and business analysis in the subsequent steps.

QUESTION 2:-

EXPLORE TABLE (SCHEMA, SAMPLE DATA)

OBJECTIVE

The objective of this step is to understand the structure, contents, and characteristics of the dataset before performing filtering, aggregation, and business analysis.

PROCEDURE

1. View Sample Records

Display the first 10 rows of the dataset:

A screenshot of a MySQL query editor window. The toolbar at the top includes icons for file operations, execution, and a dropdown menu set to 'Limit to 1000 rows'. The query editor contains the following SQL code:

```
1 • select * FROM superstore_sales
2 limit 10;
3
```

Explanation

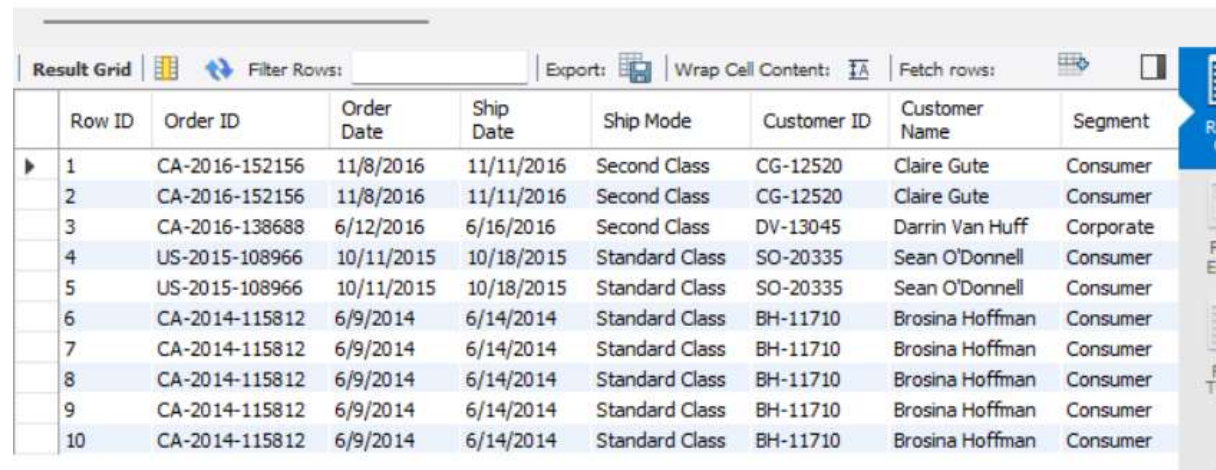
SELECT *

Retrieves all columns from the table.

LIMIT 10

Displays only the first 10 records.

OUTPUT:-



The screenshot shows a data grid interface with a toolbar at the top. The toolbar includes a 'Result Grid' button, a 'Filter Rows' input field, an 'Export' button, a 'Wrap Cell Content' button, and a 'Fetch rows' button. The table below has 9 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, and Segment. It displays 10 rows of data, with the first two rows having identical values for Order ID, Order Date, Ship Date, Ship Mode, and Customer ID, but different Customer Names.

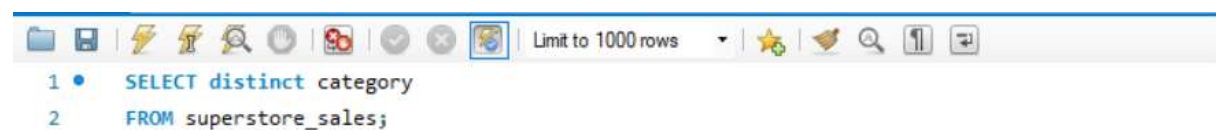
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer

Observation

The data contains order details, customer information, product categories, sales amounts, quantities, and profit values.

2. Identify Available Product Categories

Retrieve all unique product categories:



The screenshot shows a SQL query editor interface with a toolbar at the top. The toolbar includes buttons for file operations, a 'Limit to 1000 rows' dropdown, and a search icon. The query text is as follows:

```
1 • SELECT distinct category
2 FROM superstore_sales;
```

OUTPUT:-

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
category			
▶ Furniture			
Office Supplies			
Technology			

3. Identify Available Regions

Retrieve all unique sales regions:

Limit to 1000 rows
1 • SELECT distinct Region
2 FROM superstore_sales;
3

OUTPUT:-

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Region			
▶ South			
West			
Central			
East			

4. Count Total Records

Determine the total number of records in the dataset.

Query 1 x
Limit to 1000 rows
1 • select COUNT(*) AS Total_Rows
2 FROM superstore_sales;
3

OUTPUT:-

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
Total_Rows			
▶ 9694			

Observation

The dataset contains 9,694 records available for analysis.

Result

The dataset was explored successfully. The table structure, total records, unique categories, regions, and customer count were identified.

Conclusion

Dataset exploration provided a clear understanding of the available data and its structure. The information gathered in this step serves as the foundation for filtering, aggregation, and business analysis in the subsequent steps.

QUESTION:- 3

Apply WHERE filters (region, category, date, sales)

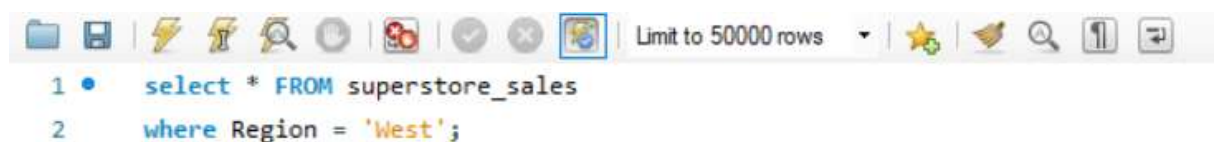
OBJECTIVE

To filter the dataset based on specific conditions such as region, category, sales amount, and date. Filtering helps in analyzing a particular subset of data relevant to business requirements.

PROCEDURE

1. Filter Records by Region

Display all orders from the West region.



The screenshot shows a SQL query editor with a toolbar at the top containing icons for file operations, execution, and search. The query text is as follows:

```
1 • select * FROM superstore_sales
2   where Region = 'West';
```

On the right side of the editor, there is a dropdown menu set to "Limit to 50000 rows" and several other utility icons.

OUTPUT:-

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10C
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	BH-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10C
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10C
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	ES-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	BH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10C
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	BH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Auzman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TEC-AC-10
64	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	OFF-BI-10C
65	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	OFF-PA-10
66	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	FUR-FU-10
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brandon Sweed	Corporate	United States	Gilbert	Arizona	85204	West	OFF-AR-10
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brandon Sweed	Corporate	United States	Gilbert	Arizona	85204	West	TEC-PH-10
87	CA-2014-136451	10/13/2014	10/16/2014	Standard Class	FW-13690	Quana Norman	Consumer	United States	San Francisco	California	94122	West	OFF-BI-10

Observation

The query returns only those sales transactions that belong to the West region.

2. Filter Records by Category

Display all orders belonging to the Technology category.

```

1 • SELECT * FROM superstore_sales
2 WHERE Category = 'Technology';
3

```

OUTPUT:-

Result Grid														Filter Rows:		Export:		Wrap Cell Contents	
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product					
▶	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH					
	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11730	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH					
	20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH					
	27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC					
	36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH					
	41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SH-20732	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	TEC-PH					
	42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Uinda Czaemsa	Corporate	United States	Naperville	Illinois	60540	Central	TEC-PH					
	45	CA-2016-118255	3/11/2016	3/13/2016	First Class	OH-16715	Odella Nelson	Corporate	United States	Eagan	Minnesota	55122	Central	TEC-AC					
	48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-AC					
	49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-PH					
	55	CA-2016-165816	12/11/2016	12/17/2016	Standard Class	JH-15265	Janet Holman	Corporate	United States	New York City	New York	10024	East	TEC-PH					
	60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East	TEC-AC					
	63	CA-2015-135540	11/24/2015	11/30/2015	Standard Class	KH-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TEC-AC					
	69	CA-2014-126376	12/5/2014	12/10/2014	Standard Class	BS-11890	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234	West	TEC-PH					
	87	CA-2017-155358	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	TEC-AC					
	91	CA-2016-129806	9/17/2016	9/22/2016	Standard Class	JS-15685	Jim Sink	Corporate	United States	Los Angeles	California	90036	West	TEC-PH					
	101	CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465	Rob Bensley	Home Office	United States	Chicago	Illinois	60610	Central	TEC-AC					
	104	US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	Lena Caccoppo	Consumer	United States	Aurora	Colorado	80013	West	TEC-AC					
	107	CA-2017-119004	11/23/2017	11/28/2017	Standard Class	JH-15250	Janet Martin	Consumer	United States	Charlotte	North Carolina	28205	South	TEC-AC					
	108	CA-2017-119004	11/23/2017	11/28/2017	Standard Class	JH-15250	Janet Martin	Consumer	United States	Charlotte	North Carolina	28205	South	TEC-PH					
110	CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19060	Pete Armstrong	Home Office	United States	Orland Park	Illinois	60462	Central	TEC-AC						
124	CA-2016-103730	6/12/2016	6/15/2016	First Class	SC-20725	Steven Carwright	Consumer	United States	Wilmington	Delaware	19805	East	TEC-PH						
131	US-2017-164147	2/2/2017	2/5/2017	First Class	DW-13585	Dorothy Wardle	Corporate	United States	Columbus	Ohio	43229	East	TEC-PH						
148	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	IF-16165	Ivett Flynn	Corporate	United States	Franklin	Wisconsin	53113	Central	TEC-PH						

Result Grid														Filter Rows:		Export:		Wrap Cell Contents:	
	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit							
	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152							
	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Korffel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568							
	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011							
	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	Ination 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	3	0	11.7741							
	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4	1097.544	7	0.2	123.4737							
	Houston	Texas	77041	Central	TEC-PH-10000486	Technology	Phones	Plantronics HL10 Handset Lifter	371.368	4	0.2	41.7564							
	Naperville	Illinois	60540	Central	TEC-PH-10004093	Technology	Phones	Panasonic Kx-TS550	147.368	4	0.2	16.5564							
	Eagan	Minnesota	55122	Central	TEC-AC-10000171	Technology	Accessories	Verbatim 25 GB 6x Blu-ray Single Layer Recordable	45.98	2	0	19.7714							
	Dover	Delaware	19901	East	TEC-AC-10002167	Technology	Accessories	Ination 8gb Micro Traveldrive Usb 2.0 Flash Drive	45	3	0	4.95							
	Dover	Delaware	19901	East	TEC-PH-10003988	Technology	Phones	LF Elite 3D Dazzle Designer Hard Case Cover, L.F...	21.8	2	0	6.304							
	New York City	New York	10024	East	TEC-PH-10002447	Technology	Phones	AT&T CL83451 4-Handset Telephone	1029.95	5	0	298.6855							
	Troy	New York	12180	East	TEC-AC-10002167	Technology	Accessories	Ination 8gb Micro Traveldrive Usb 2.0 Flash Drive	30	2	0	3.3							
	Los Angeles	California	90004	West	TEC-AC-10004633	Technology	Accessories	Verbatim 25 GB 6x Blu-ray Single Layer Recordable	13.98	2	0	6.1512							
	Gilbert	Arizona	85234	West	TEC-PH-10002726	Technology	Phones	netTALK DUO VoIP Telephone Service	167.968	4	0.2	62.988							
	Rochester	Minnesota	55901	Central	TEC-AC-10001998	Technology	Accessories	Logitech LS21 Speaker System - PC Multimedia	19.99	1	0	6.7966							
	Los Angeles	California	90036	West	TEC-PH-10004093	Technology	Phones	Panasonic Kx-TS550	73.584	2	0.2	8.2782							
	Chicago	Illinois	60610	Central	TEC-AC-10001767	Technology	Accessories	SanDisk Ultra 64 GB MicroSDHC Class 10 Memor...	95.976	3	0.2	-10.7973							
	Aurora	Colorado	80013	West	TEC-AC-10001552	Technology	Accessories	Logitech K350 2.4Ghz Wireless Keyboard	238.896	6	0.2	-26.8758							
	Charlotte	North Carolina	28205	South	TEC-AC-10003499	Technology	Accessories	Memorex Mini Travel Drive 8 GB USB 2.0 Flash D...	74.112	8	0.2	17.6016							
	Charlotte	North Carolina	28205	South	TEC-PH-10002844	Technology	Phones	Speck Products Candyshell Flip Case	27.992	1	0.2	2.0994							
Orland Park	Illinois	60462	Central	TEC-AC-10000844	Technology	Accessories	Logitech Gaming G510s - Keyboard	339.96	5	0.2	67.992								
Wilmington	Delaware	19805	East	TEC-PH-10003875	Technology	Phones	KLD Oscar II Style Snap-on Ultra Thin Side Flip S...	68.04	7	0	19.7316								
Columbus	Ohio	43229	East	TEC-PH-10002293	Technology	Phones	Anker 36W 4-Port USB Wall Charger Travel Pow...	59.97	5	0.4	-11.994								
Franklin	Wisconsin	53113	Central	TEC-PH-10000215	Technology	Phones	Plantronics Cordless Phone Headset with 3-Inch...	98.45	11	0	0.4	103.8015							

Observation

Technology products can now be analyzed separately to evaluate their sales and profit performance.

3. Filter High Sales Orders

Display orders where sales are greater than 500.


```

1 • SELECT * FROM superstore_sales
2   WHERE Sales > 500;
3
4

```

OUTPUT:-

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	PUR
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	PUR
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	PUR
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC
17	CA-2014-105893	11/11/2014	11/18/2014	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	PUR
28	US-2015-130630	9/17/2015	9/21/2015	Standard Class	TS-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	PUR
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC
39	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SH-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	PUR
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East	TEC
68	CA-2014-106326	12/25/2014	12/30/2014	Standard Class	BS-11590	Brendan Sneed	Corporate	United States	Gilbert	Arizona	85234	West	OFF
73	US-2015-134026	4/26/2015	5/2/2015	Standard Class	JE-15745	Joel Eaton	Consumer	United States	Memphis	Tennessee	38109	South	PUR
118	CA-2015-110457	3/2/2015	3/5/2015	Standard Class	DK-13090	Dave Kipp	Consumer	United States	Seattle	Washington	98103	West	PUR
125	US-2014-152030	12/26/2014	12/28/2014	Second Class	AD-10180	Alan Dominguez	Home Office	United States	Houston	Texas	77041	Central	PUR
126	US-2014-134614	9/20/2014	9/25/2014	Standard Class	PF-15165	Philip Fox	Consumer	United States	Bloomington	Illinois	61701	Central	PUR
145	CA-2017-155376	12/22/2017	12/27/2017	Standard Class	SG-20080	Sandra Glasco	Consumer	United States	Independence	Missouri	64055	Central	OFF
146	CA-2015-110744	9/7/2015	9/12/2015	Standard Class	HA-14920	Helen Andreada	Consumer	United States	Pasadena	California	91104	West	OFF
150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16365	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central	PUR
160	CA-2016-114104	11/20/2016	11/24/2016	Standard Class	NP-18670	Nora Paige	Consumer	United States	Edmond	Oklahoma	73034	Central	TEC
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	TEC
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	PUR
183	CA-2014-138274	11/9/2014	11/24/2014	Second Class	RM-19675	Robert Marley	Home Office	United States	Monroe	Louisiana	71203	South	TEC
190	CA-2015-107381	10/17/2015	10/14/2015	First Class	MB-17420	Mark Parker	Home Office	United States	New York City	New York	10019	East	PUR

State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Florida	32216	South	TEC-MA-10002412	Technology	Machines	Canon TelePresence System EX90 Videoconferen...	22638.48	6	0.5	-1811.0784
Indiana	47905	Central	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	17499.95	5	0	8399.975
Washington	98115	West	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	13999.96	4	0	6719.9808
New York	10024	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	11199.968	4	0.2	3919.9888
Delaware	19711	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	10499.97	3	0	5039.9856
Michigan	48203	Central	OFF-BI-10000545	Office Supplies	Binders	GBC IstaMaster 500 Manual ProClick Binding System	9892.74	13	0	4946.37
Minnesota	55407	Central	OFF-BI-10001120	Office Supplies	Binders	Bico EPK-21 Electric Binding System	9449.95	5	0	4630.4755
New Jersey	8701	East	TEC-MA-10001047	Technology	Machines	3D Systems Cube Printer, 2nd Generation, Mag...	9099.93	7	0	2385.9818
Virginia	22204	South	TEC-MA-10001127	Technology	Machines	HP DesignJet T320 Inkjet Large Format Printer ...	8749.95	5	0	2799.984
Pennsylvania	19120	East	TEC-CO-10004722	Technology	Copiers	Canon imageCLASS 2200 Advanced Copier	8399.976	4	0.4	1119.9968
California	94122	West	OFF-SU-10000151	Office Supplies	Supplies	High Speed Automatic Electric Letter Opener	8187.65	5	0	327.506
Texas	78207	Central	TEC-MA-10000822	Technology	Machines	Lexmark MX611dhe Monochrome Laser Printer	8159.952	8	0.4	-1359.992
North Carolina	27217	South	TEC-MA-10004125	Technology	Machines	Cubify CubeX 3D Printer Triple Head Print	7999.98	4	0.5	-3839.9904
New York	10024	East	TEC-MA-10001127	Technology	Machines	HP DesignJet T320 Inkjet Large Format Printer ...	6999.96	4	0	2239.9872
Georgia	30318	South	OFF-BI-10003527	Office Supplies	Binders	Fellowes PB500 Electric Punch Plastic Comb Bind...	6354.95	5	0	3177.475
Michigan	48201	Central	OFF-BI-10004995	Office Supplies	Binders	GBC DocuBind P400 Electric Binding System	5443.96	4	0	2524.2216
Rhode Island	2908	East	TEC-CO-10001449	Technology	Copiers	Hewlett Packard Laser Jet 3310 Copier	5399.91	9	0	2591.9568
New York	10035	East	TEC-MA-10001047	Technology	Machines	3D Systems Cube Printer, 2nd Generation, Mag...	5199.96	4	0	1351.9896
California	95823	West	OFF-BI-10002027	Office Supplies	Binders	Fellowes PB500 Electric Punch Plastic Comb Bind...	5003.96	5	0.2	1906.485
California	94109	West	OFF-SU-10000151	Office Supplies	Supplies	High Speed Automatic Electric Letter Opener	4912.93	3	0	196.5036
New York	10701	East	TEC-MA-10003979	Technology	Machines	Ativa V4110MDD Micro-Cut Shredder	4899.93	7	0	2400.9657
Missouri	65807	Central	TEC-CO-10003763	Technology	Copiers	Canon PC1060 Personal Laser Copier	4899.93	7	0	2302.9671
California	94110	West	TEC-MA-10000418	Technology	Machines	Cubify CubeX 3D Printer Double Head Print	4799.984	2	0.2	359.9988
Pennsylvania	19114	East	OFF-SU-10003881	Office Supplies	Supplies	Holten Yale Charlesstream Electric Letter Op...	4663.756	7	0.7	-1049.7456

4. Filter Records by Date

Display orders placed on or after 1 January 2017.

```

1 • SELECT * FROM superstore_sales
2   WHERE STR_TO_DATE('Order Date', '%c/%e/%Y') > '2017-01-01';
3
4

```

OUTPUT:-

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Prod
13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-I
24	US-2017-156909	7/16/2017	7/18/2017	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-I
35	CA-2017-107727	10/19/2017	10/23/2017	Second Class	MA-17560	Matt Abelman	Home Office	United States	Houston	Texas	77095	Central	OFF-I
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	United States	Naperville	Illinois	60540	Central	TEC-I
44	CA-2017-139619	9/19/2017	9/23/2017	Standard Class	ES-14080	Erin Smith	Corporate	United States	Melbourne	Florida	32935	South	OFF-I
72	CA-2017-114440	9/14/2017	9/17/2017	Second Class	TB-21520	Tracy Blumstein	Consumer	United States	Jackson	Michigan	48201	Central	OFF-I
76	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OFF-I
78	US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OFF-I
85	US-2017-119662	11/13/2017	11/16/2017	First Class	CS-12400	Christopher Schild	Home Office	United States	Chicago	Illinois	60623	Central	OFF-I
86	CA-2017-140088	5/28/2017	5/30/2017	Second Class	PO-18865	Patrick O'Donnell	Consumer	United States	Columbia	South Carolina	29203	South	FUR-I
87	CA-2017-155558	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	TEC-I
88	CA-2017-155558	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	OFF-I
96	US-2017-109484	11/6/2017	11/12/2017	Standard Class	RB-19705	Roger Barco	Home Office	United States	Portland	Oregon	97206	West	OFF-I
97	CA-2017-161018	11/9/2017	11/11/2017	Second Class	PH-18775	Parthena Horis	Home Office	United States	New York ...	New York	10009	East	FUR-I

City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	15.552	3	0.2	5.4432
Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs	Global Deluxe Stacking Chair, Gray	71.372	2	0.3	-1.0196
Houston	Texas	77095	Central	OFF-PA-10000249	Office Supplies	Paper	Easy-staple paper	29.472	3	0.2	9.9468
Naperville	Illinois	60540	Central	TEC-PH-10004093	Technology	Phones	Panasonic KX-TS550	147.168	4	0.2	36.5564
Melbourne	Florida	32935	South	OFF-ST-10003282	Office Supplies	Storage	Advantus 10-Drawer Portable Organizer, Chro...	95.616	2	0.2	9.5616
Jackson	Michigan	48201	Central	OFF-PA-10004675	Office Supplies	Paper	Telephone Message Books with Fax/Mobile Secti...	19.05	3	0	8.763
Houston	Texas	77041	Central	OFF-BI-10004182	Office Supplies	Binders	Economy Binders	1.248	3	0.8	-1.9344
Houston	Texas	77041	Central	OFF-ST-10000618	Office Supplies	Storage	Simplify Personal File, Black Granite, 15in x 6-1...	27.24	3	0.2	2.724
Chicago	Illinois	60623	Central	OFF-ST-10003656	Office Supplies	Storage	Safco Industrial Wire Shelving	230.376	3	0.2	-48.9549
Columbia	South Carolina	29203	South	FUR-CH-10000863	Furniture	Chairs	Novimes Swivel Fabric Task Chair	301.96	2	0	33.2156
Rochester	Minnesota	55901	Central	TEC-AC-10001998	Technology	Accessories	Logitech LS21 Speaker System - PC Multimedia ...	19.99	1	0	6.7966
Rochester	Minnesota	55901	Central	OFF-LA-10000134	Office Supplies	Labels	Avery 511	6.36	2	0	2.9568
Portland	Oregon	97206	West	OFF-BI-10004738	Office Supplies	Binders	Flexible Leather-Look Classic Collection Ring Br...	5.682	1	0.7	-3.788
New York ...	New York	10009	East	FUR-FU-10000629	Furniture	Furnishings	9-3/4 Diameter Round Wall Clock	96.53	7	0	40.5426

Observation

This filter helps analyze recent sales trends and customer purchasing behavior.

5. Apply Multiple Conditions

Display Furniture category orders from the East region with sales greater than 300.

```

1 • SELECT * FROM superstore_sales
2 WHERE Region = 'East'
3 AND Category = 'Furniture'
4 AND Sales > 300;
5
6
7

```

OUTPUT:-

Result Grid												
Filter Rows												
Export: Wrap Cell Contents												
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East
58	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East
178	US-2015-101513	11/21/2015	11/23/2015	Second Class	JE-15745	Joel Eaton	Consumer	United States	Newark	Ohio	43055	East
190	CA-2015-802281	10/12/2015	10/14/2015	First Class	MP-17470	Mark Packer	Home Office	United States	New York City	New York	10035	East
193	CA-2015-802281	10/12/2015	10/14/2015	First Class	MP-17470	Mark Packer	Home Office	United States	New York City	New York	10035	East
214	CA-2015-146262	1/2/2015	1/9/2015	Standard Class	VW-21775	Victoria Wilson	Corporate	United States	Medina	Ohio	44256	East
329	US-2016-141544	8/30/2015	9/1/2016	First Class	PO-18850	Pwinkl O'Brien	Consumer	United States	Philadelphia	Pennsylvania	19143	East
353	CA-2016-138520	4/8/2016	4/13/2016	Standard Class	JE-15505	Jeremy Lonsdale	Consumer	United States	New York City	New York	10035	East
363	CA-2017-144904	9/25/2017	10/1/2017	Standard Class	KW-16435	Katrina Wilman	Consumer	United States	New York City	New York	10009	East
370	CA-2016-155536	10/21/2016	10/21/2016	Same Day	MC-17905	Michael Kennedy	Corporate	United States	Manchester	Connecticut	6040	East
378	US-2017-134461	8/27/2017	9/1/2017	Standard Class	AR-10405	Allen Rosenblatt	Corporate	United States	Franklin	Massachusetts	2038	East
709	CA-2014-112158	12/2/2014	12/4/2014	First Class	DP-13165	David Philippe	Consumer	United States	New York City	New York	10035	East
729	US-2016-150961	12/3/2016	12/6/2016	First Class	EG-13900	Emily Grady	Consumer	United States	Oceanside	New York	11572	East
731	US-2016-150961	12/3/2016	12/6/2016	First Class	EG-13900	Emily Grady	Consumer	United States	Oceanside	New York	11572	East

Result Grid												
Filter Rows												
Export: Wrap Cell Contents												
	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
▶	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royal...	3083.43	7	0.5	-1665.0522
	Troy	New York	12180	East	FUR-CH-10003968	Furniture	Chairs	Novimex Turbo Task Chair	319.41	5	0.1	7.098
	Newark	Ohio	43055	East	FUR-CH-10004698	Furniture	Chairs	Padded Folding Chairs, Black, 4/Carton	396.802	7	0.3	-11.3372
	New York City	New York	10035	East	FUR-BO-10002613	Furniture	Bookcases	Atlantic Metals Mobile 4-Shelf Bookcases, Custo...	899.136	4	0.2	112.392
	New York City	New York	10035	East	FUR-BO-10002545	Furniture	Bookcases	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...	626.352	3	0.2	46.9764
	Medina	Ohio	44256	East	FUR-BO-10004695	Furniture	Bookcases	O'Sullivan 2-Door Barrister Bookcase in Odesa ...	452.45	5	0.5	-244.323
	Philadelphia	Pennsylvania	19143	East	FUR-CH-10003312	Furniture	Chairs	Hon 2090 "Pillow Soft" Series Mid Back Swivel/Tk...	796.744	4	0.3	-258.5016
	New York City	New York	10035	East	FUR-BO-10002268	Furniture	Bookcases	Sauder Barrister Bookcases	388.704	6	0.2	-4.8588
	New York City	New York	10009	East	FUR-CH-10000785	Furniture	Chairs	Global Ergonomic Managers Chair	488.646	3	0.1	86.8704
	Manchester	Connecticut	6040	East	FUR-BO-10002545	Furniture	Bookcases	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...	1043.92	4	0	271.4192
	Franklin	Massachusetts	2038	East	FUR-TA-10004915	Furniture	Tables	Office Impressions End Table, 20-1/2"H x 24"...	1488.424	7	0.3	-297.6848
	New York City	New York	10035	East	FUR-BO-10003272	Furniture	Bookcases	O'Sullivan Living Dimensions 5-Shelf Bookcases	883.92	5	0.2	-120.49
	Oceanside	New York	11572	East	FUR-TA-10002228	Furniture	Tables	Bevis Traditional Conference Table Top, Plinth B...	400.032	2	0.4	-153.3456
	Oceanside	New York	11572	East	FUR-CH-10002965	Furniture	Chairs	Global Leather Highback Executive Chair with P...	542.646	3	0.1	102.498

Observation

Combining multiple filters enables focused analysis of a specific market segment.

Result

- Total sales were calculated region-wise.
- Average profit was calculated category-wise.
- Total quantity sold was determined for each category.
- Region-wise order counts were identified.

Conclusion

The GROUP BY clause was successfully used to aggregate and summarize sales data. These results provide valuable business insights regarding sales performance, profitability, and customer activity across different regions and categories.

QUESTION :- 4

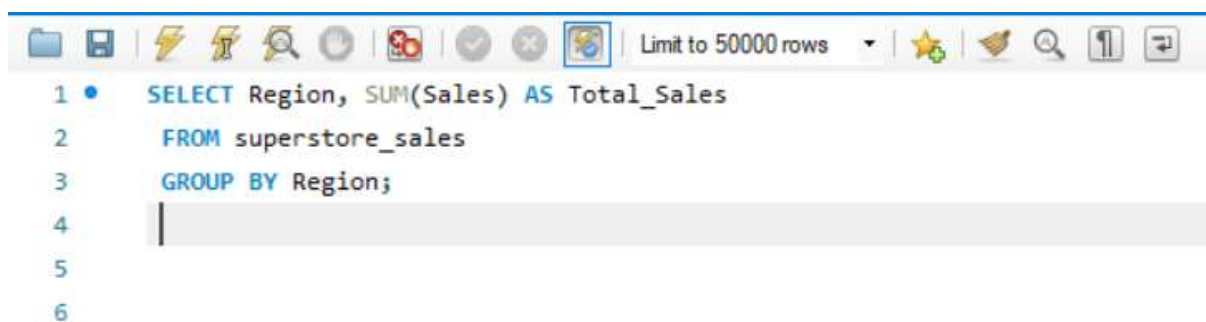
Use GROUP BY for Aggregations (Sales, Quantity, and Averages)

OBJECTIVE

To summarize the sales data using the GROUP BY clause and calculate total sales, total quantity sold, and average profit across different business categories and regions.

1. Calculate Total Sales by Region

To calculate the total revenue generated from each region.

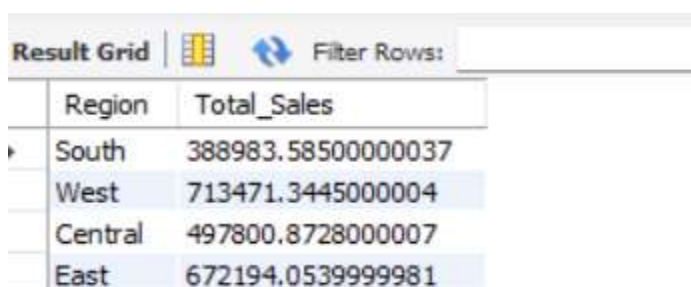


```
1 • SELECT Region, SUM(Sales) AS Total_Sales
2   FROM superstore_sales
3  GROUP BY Region;
4
5
6
```

Explanation

- SUM(Sales) calculates the total sales amount.
- GROUP BY Region groups all records belonging to the same region.
- The query returns one row for each region.

OUTPUT:-



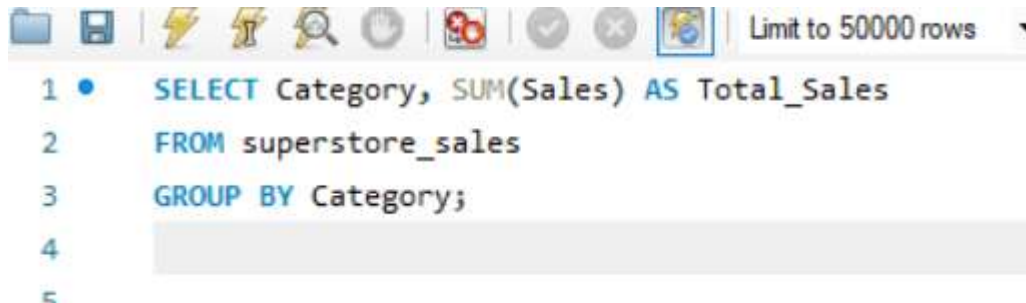
Region	Total_Sales
South	388983.58500000037
West	713471.34450000004
Central	497800.87280000007
East	672194.0539999981

Observation

This query helps identify which region contributes the highest sales revenue.

2. Total Sales by Category

To determine the total sales generated by each product category.

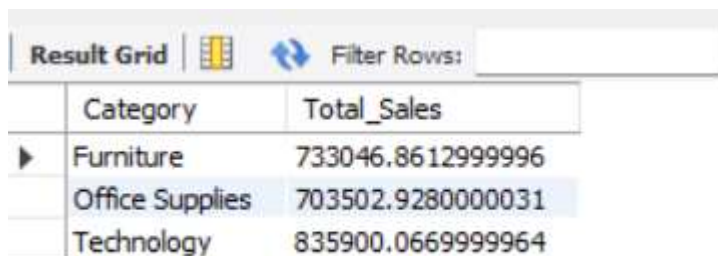


```
1 • SELECT Category, SUM(Sales) AS Total_Sales
2 FROM superstore_sales
3 GROUP BY Category;
```

Explanation

- SUM(Sales) adds all sales values.
- GROUP BY Category groups records according to product categories.
- The query provides total sales for each category.

OUTPUT:-



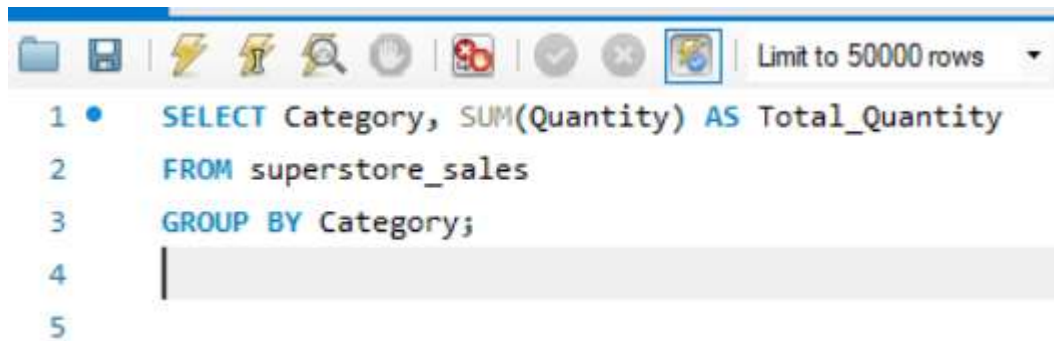
Category	Total_Sales
Furniture	733046.8612999996
Office Supplies	703502.9280000031
Technology	835900.0669999964

Observation

This analysis shows which category generates the highest revenue.

3. Total Quantity Sold by Category

To determine the total number of products sold in each category.

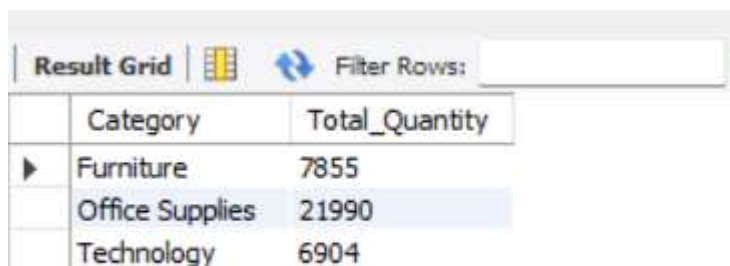


```
1 • SELECT Category, SUM(Quantity) AS Total_Quantity
2 FROM superstore_sales
3 GROUP BY Category;
4
5
```

Explanation

- SUM(Quantity) adds all product quantities sold.
- GROUP BY Category groups records category-wise.
- The query calculates total units sold per category.

OUTPUT:-



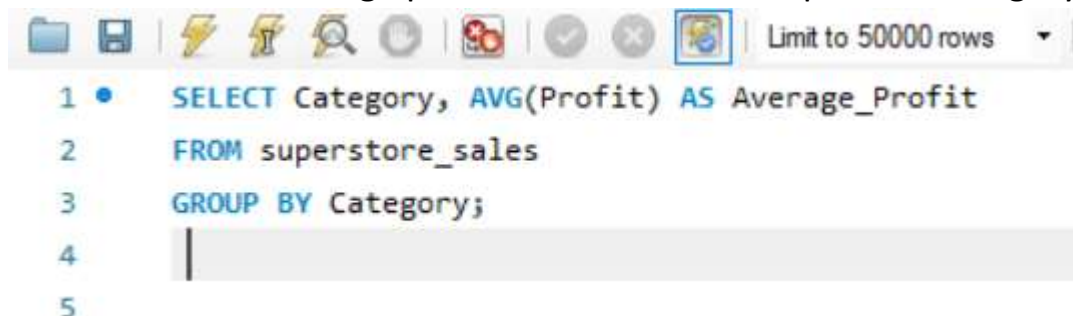
	Category	Total_Quantity
▶	Furniture	7855
	Office Supplies	21990
	Technology	6904

Observation

This query identifies the category with the highest product demand.

4. Average Profit by Category

To calculate the average profit earned from each product category.



```
1 • SELECT Category, AVG(Profit) AS Average_Profit
2 FROM superstore_sales
3 GROUP BY Category;
4
5
```

Explanation

- AVG(Profit) calculates the average profit.

- GROUP BY Category creates separate groups for each category.
- The query compares profitability across categories.

OUTPUT:-

Result Grid		Filter Rows:
	Category	Average_Profit
▶	Furniture	8.187449951783982
	Office Supplies	20.84239515654733
	Technology	79.05769255029902

Observation

The category with the highest average profit is considered the most profitable on average.

Result

- Total sales were calculated for each region.
- Total sales were calculated for each category.
- Total quantity sold was determined category-wise.
- Average profit was calculated for each category.

Conclusion

The GROUP BY clause was successfully used to aggregate data and generate meaningful business insights. The analysis revealed sales performance, product demand, and profitability across different regions and categories, enabling better business decision-making.

QUESTION:-5

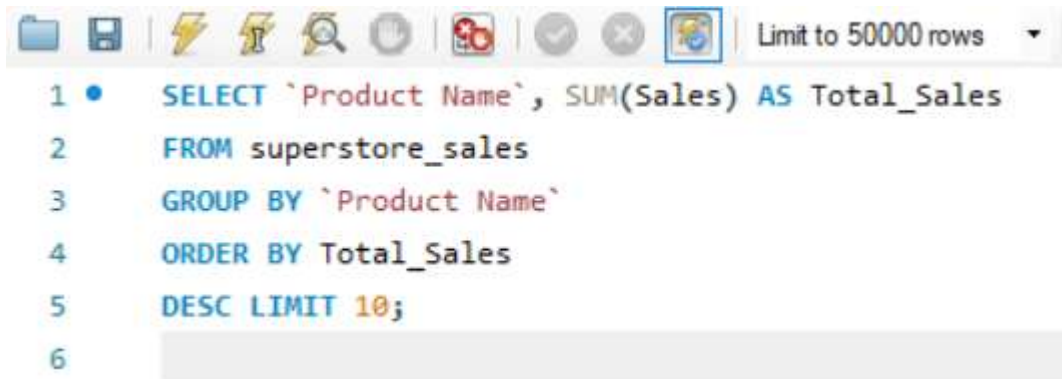
SORT AND LIMIT RESULTS (TOP PRODUCTS AND TOP CATEGORIES)

OBJECTIVE

The objective of this step is to identify the top-performing products and categories by sorting the aggregated results and limiting the output to the highest-performing records. This helps in recognizing the major contributors to sales and revenue.

1. Top 10 Products by Sales

To identify the products that generated the highest sales revenue.



```
1 • SELECT `Product Name`, SUM(Sales) AS Total_Sales
2 FROM superstore_sales
3 GROUP BY `Product Name`
4 ORDER BY Total_Sales
5 DESC LIMIT 10;
6
```

Explanation

- SUM(Sales) calculates total sales for each product.
- GROUP BY Product Name groups records product-wise.
- ORDER BY Total_Sales DESC sorts products from highest to lowest sales.
- LIMIT 10 displays only the top 10 products.

OUTPUT:-

Result Grid



Filter Rows:

Export:


Wrap Cell Content:

	Product Name	Total_Sales
▶	Canon imageCLASS 2200 Advanced Copier	61599.824
	Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.384
	Cisco TelePresence System EX90 Videoconferen...	22638.48
	HON 5400 Series Task Chairs for Big and Tall	21870.576
	GBC DocuBind TL300 Electric Binding System	19823.479000000003
	GBC Ibimaster 500 Manual ProClick Binding System	19024.5
	Hewlett Packard LaserJet 3310 Copier	18839.686
	HP Designjet T520 Inkjet Large Format Printer -...	18374.895
	GBC DocuBind P400 Electric Binding System	17965.068
	High Speed Automatic Electric Letter Opener	17030.311999999998

Observation

This query identifies the best-selling products that contribute the most revenue to the business.

2. Top 5 Categories by Sales

To determine which product categories generate the highest sales.

```

1 • SELECT Category,
2     SUM(Sales) AS Total_Sales
3     FROM superstore_sales
4     GROUP BY Category
5     ORDER BY Total_Sales DESC
6     LIMIT 5;
7

```

Explanation

- Sales are aggregated category-wise.
- Results are sorted in descending order of total sales.
- Only the top categories are displayed.

OUTPUT:-

Result Grid		
	Category	Total_Sales
▶	Technology	835900.0669999964
	Furniture	733046.8612999996
	Office Supplies	703502.9280000031

Observation

The category at the top of the list contributes the highest revenue.

3. Top 10 Customers by Sales

To identify customers who generated the highest sales value.

```

1 • SELECT `Customer Name`,
2     SUM(Sales) AS Total_Sales
3     FROM superstore_sales
4     GROUP BY `Customer Name`
5     ORDER BY Total_Sales
6     DESC LIMIT 10;
7

```

Explanation

- Groups records by customer.
- Calculates total sales for each customer.
- Sorts customers based on total sales.
- Displays the top 10 customers.

OUTPUT:-

Result Grid			Filter Rows:
	Customer Name	Total_Sales	
▶	Sean Miller	25043.05	
	Tamara Chand	19017.847999999998	
	Raymond Buch	15117.339	
	Tom Ashbrook	14595.62	
	Adrian Barton	14355.610999999997	
	Sanjit Chand	14142.333999999999	
	Ken Lonsdale	14071.917	
	Hunter Lopez	12873.297999999999	
	Sanjit Engle	12209.438000000002	
	Christopher Conant	12129.072	

Observation

This analysis helps identify the most valuable customers for the business.

4. Top 10 Products by Quantity Sold

To identify the most frequently purchased products.

```

1 • SELECT `Product Name`,
2     SUM(Quantity) AS Total_Quantity
3     FROM superstore_sales
4     GROUP BY `Product Name`
5     ORDER BY Total_Quantity
6     DESC LIMIT 10;
7

```

Explanation

- SUM(Quantity) calculates total units sold.
- Products are sorted from highest to lowest quantity sold.
- The top 10 products are displayed.

OUTPUT:-

Result Grid			Filter Rows:	Export:	W
	Product Name	Total_Quantity			
▶	Staples	215			
	Staple envelope	170			
	Easy-staple paper	150			
	Staples in misc. colors	86			
	KI Adjustable-Height Table	74			
	Storex Dura Pro Binders	71			
	Avery Non-Stick Binders	71			
	GBC Premium Transparent Covers with Diagonal...	67			
	Situations Contoured Folding Chairs, 4/Set	64			
	Staple-based wall hangings	62			

Observation

This query highlights the products with the highest customer demand.

Result

- Top-selling products were identified.
- Top-performing categories were determined.
- Highest-value customers were identified.
- Most frequently purchased products were analyzed.

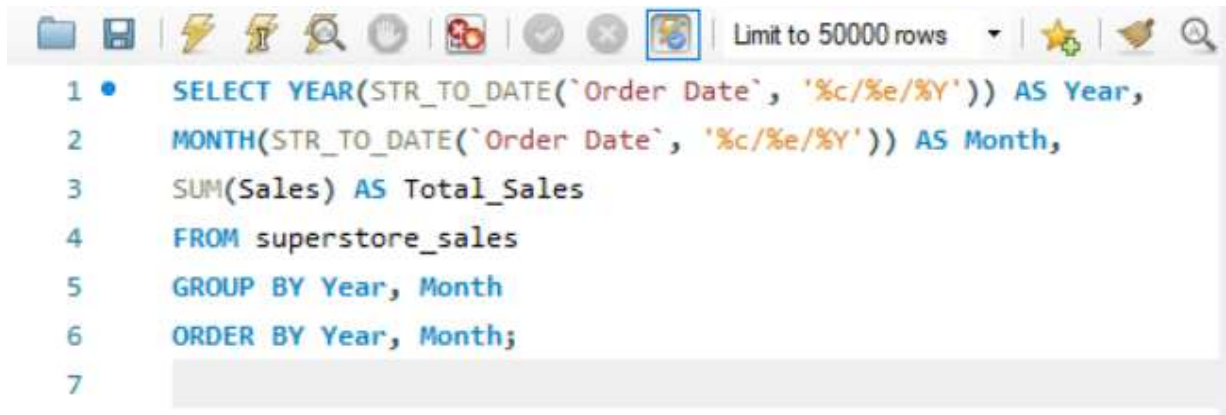
Conclusion

Sorting and limiting results using ORDER BY and LIMIT helped identify the most important products, categories, and customers. These insights can support inventory planning, marketing strategies, and business decision-making.

QUESTION:- 6

SOLVE BUSINESS USE CASES (MONTHLY TRENDS, TOP CUSTOMERS, AND DUPLICATE RECORDS)

OBJECTIVE



```
1 • SELECT YEAR(STR_TO_DATE(`Order Date`, '%c/%e/%Y')) AS Year,  
2     MONTH(STR_TO_DATE(`Order Date`, '%c/%e/%Y')) AS Month,  
3     SUM(Sales) AS Total_Sales  
4     FROM superstore_sales  
5     GROUP BY Year, Month  
6     ORDER BY Year, Month;  
7
```

The objective of this step is to solve real-world business use cases using SQL queries. The analysis focuses on monthly sales trends, identifying top customers, and detecting duplicate records in the dataset.

1. Monthly Sales Trend Analysis

To analyze how sales vary over different months and identify seasonal trends.

Explanation

- STR_TO_DATE() converts text dates into date format.
- YEAR() extracts the year.
- MONTH() extracts the month.
- SUM(Sales) calculates monthly sales.
- Results are grouped and sorted chronologically.

OUTPUT:-

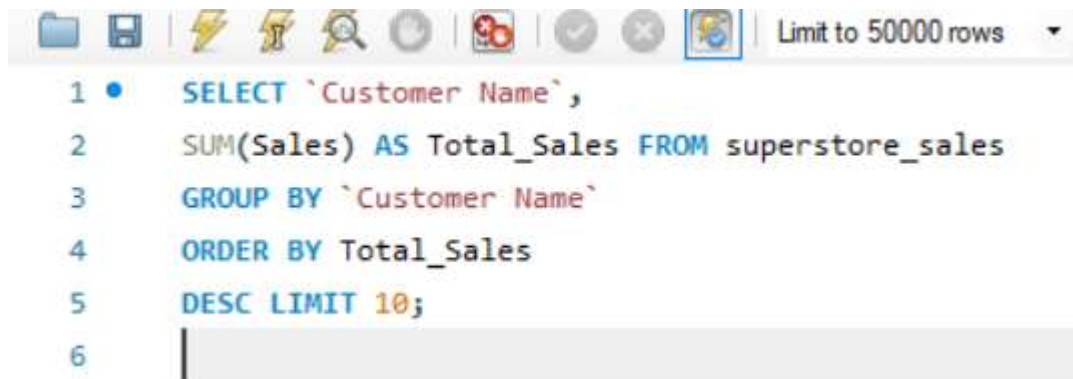
Result Grid				Filter Rows:		
	Year	Month	Total_Sales			
▶	2014	1	14161.348999999998	2016	1	18432.590999999993
	2014	2	4119.816	2016	2	22706.415
	2014	3	55526.199000000002	2016	3	50832.549000000002
	2014	4	28139.560999999994	2016	4	38587.140999999974
	2014	5	23634.667	2016	5	56457.319999999985
	2014	6	34508.995600000003	2016	6	39628.19199999998
	2014	7	33500.872999999999	2016	7	39108.00799999997
	2014	8	27603.512499999999	2016	8	31014.4023
	2014	9	81496.80679999998	2016	9	71848.25089999994
	2014	10	31394.940999999988	2016	10	58120.423000000001
	2014	11	78297.24069999997	2016	11	78454.90179999999
	2014	12	69379.83649999999	2016	12	96075.067000000002
	2015	1	18085.115599999994	2017	1	43860.286000000001
	2015	2	11924.271999999999	2017	2	20262.322399999997
	2015	3	38621.291999999994	2017	3	58739.223799999985
	2015	4	32640.482499999987	2017	4	36020.173100000001
	2015	5	29325.9705000000003	2017	5	44095.3522
	2015	6	24659.6840000000005	2017	6	52242.5617
	2015	7	28524.520999999999	2017	7	44490.4480000000004
	2015	8	36380.928199999995	2017	8	62643.40799999998
	2015	9	63704.302000000004	2017	9	86487.306000000004
	2015	10	31382.155500000001	2017	10	77542.48319999997
	2015	11	74699.033500000005	2017	11	117383.381000000004
	2015	12	74478.479200000007	2017	12	81227.6158

Observation

The query helps identify months with high and low sales performance and reveals seasonal sales patterns.

2. Top 10 Customers by Sales

To identify the customers contributing the highest revenue.

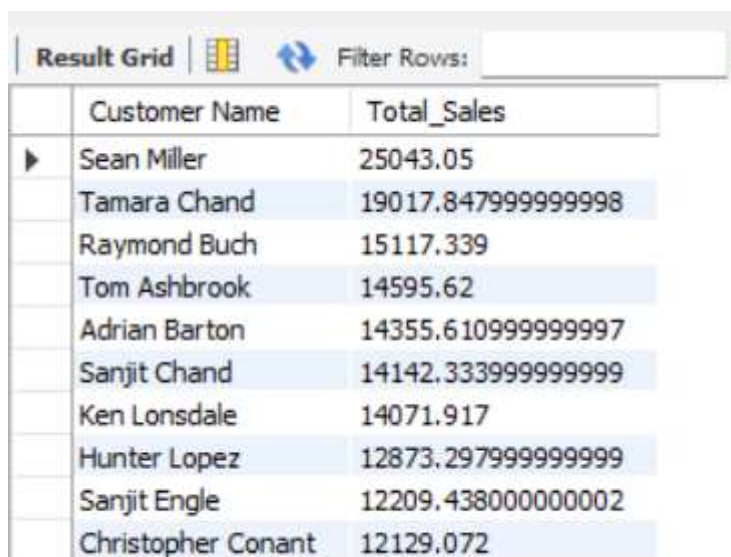


```
1 • SELECT `Customer Name`,  
2     SUM(Sales) AS Total_Sales FROM superstore_sales  
3     GROUP BY `Customer Name`  
4     ORDER BY Total_Sales  
5     DESC LIMIT 10;  
6
```

Explanation

- Groups records customer-wise.
- Calculates total sales for each customer.
- Sorts customers by highest sales.
- Displays the top 10 customers.

OUTPUT:-



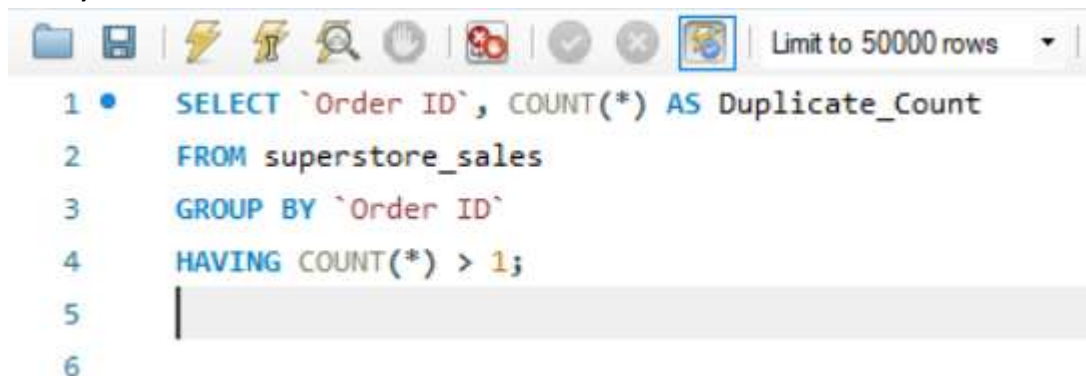
	Customer Name	Total_Sales
▶	Sean Miller	25043.05
	Tamara Chand	19017.847999999998
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14355.610999999997
	Sanjit Chand	14142.333999999999
	Ken Lonsdale	14071.917
	Hunter Lopez	12873.297999999999
	Sanjit Engle	12209.438000000002
	Christopher Conant	12129.072

Observation

The top customers contribute a significant portion of the company's revenue and can be targeted for customer retention programs.

3. Identify Duplicate Records

To detect duplicate order records that may affect data quality and analysis.



```
1 • SELECT `Order ID`, COUNT(*) AS Duplicate_Count
2 FROM superstore_sales
3 GROUP BY `Order ID`
4 HAVING COUNT(*) > 1;
5
6
```

Explanation

- Groups records by Order ID.
- Counts occurrences of each order.
- Displays only those orders appearing more than once.
- **COUNT(*)**
 - Counts the number of occurrences of each Order ID.
- **GROUP BY Order ID**
 - Groups records according to Order ID.
- **HAVING COUNT(*) > 1**
 - Displays only those Order IDs that appear more than once.

OUTPUT:-

Result Grid			Result Grid		
		Filter Rows:			Filter Rows:
	Order ID	Duplicate_Count		Order ID	Duplicate_Count
▶	CA-2016-152156	2		CA-2016-127208	2
	US-2015-108966	2		CA-2014-139451	2
	CA-2014-115812	7		CA-2017-155558	2
	US-2015-118983	2		CA-2016-109806	3
	CA-2014-143336	3		CA-2015-149587	3
	CA-2016-137330	2		CA-2016-158568	3
	CA-2016-121755	2		US-2015-156867	2
	US-2015-150630	6		CA-2017-119004	3
	CA-2016-117590	2		CA-2016-128867	2
	CA-2015-117415	4		CA-2014-115259	4
	CA-2016-118255	2		CA-2016-103730	5
	CA-2016-169194	2		US-2017-107272	2
	CA-2015-115742	4		US-2016-125969	2
	CA-2016-105816	2		US-2017-164147	3
	CA-2016-111682	7		CA-2016-145583	7
	CA-2015-135545	4		CA-2017-106180	3
	CA-2014-106376	2		CA-2016-114489	4
	US-2015-134026	3		CA-2016-158834	2
	US-2017-118038	2		CA-2015-124919	3

Observation

Duplicate records, if present, should be reviewed to ensure accurate reporting and analysis.

Result

- Monthly sales trends were analyzed successfully.
- Top-performing customers were identified.
- Duplicate records were detected and reviewed.

Conclusion

Business use cases were successfully solved using SQL queries. The analysis provided insights into sales trends, customer contribution, and data quality. These findings support better business decision-making and improve the reliability of future analysis.

QUESTION:- 7

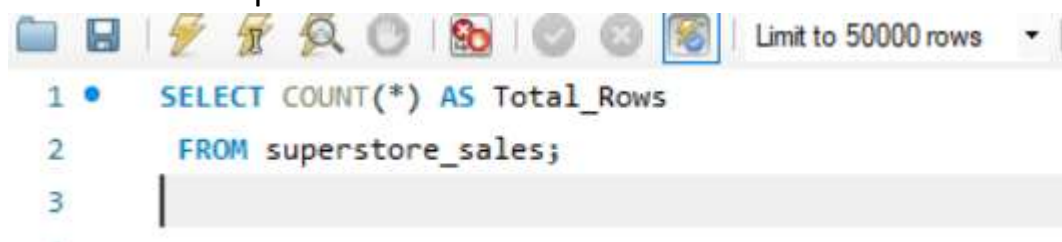
VALIDATE RESULTS (ROW COUNTS AND DATA QUALITY CHECKS)

OBJECTIVE

The objective of this step is to validate the dataset and ensure data quality before drawing business conclusions. Data validation helps identify missing values, duplicate records, invalid entries, and inconsistencies that could affect the accuracy of analysis.

1. Verify Total Number of Records

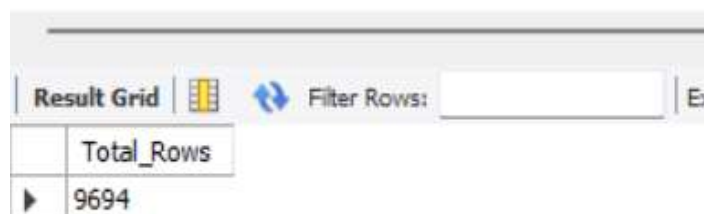
To confirm that the dataset has been imported successfully and contains the expected number of records.



Explanation

COUNT(*)

- . Counts all rows present in the dataset.
- . This query helps verify whether the data import process was completed successfully



The screenshot shows a 'Result Grid' window with a toolbar at the top. The grid contains the following data:

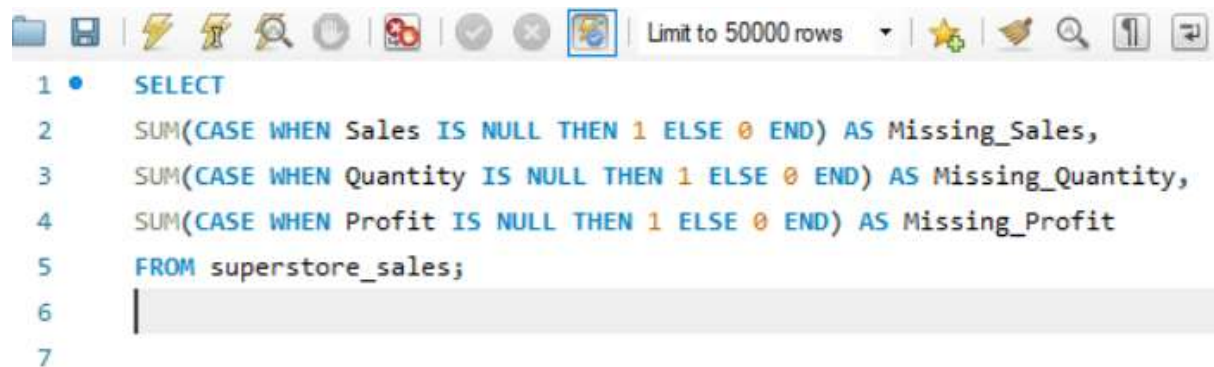
	Total_Rows
▶	9694

Observation

The dataset contains 9,694 records, indicating that the import process was completed successfully.

2 Check for Missing Values

Missing values can impact analysis and lead to incorrect business decisions. This query identifies null values in important columns.



```
1 • SELECT
2     SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS Missing_Sales,
3     SUM(CASE WHEN Quantity IS NULL THEN 1 ELSE 0 END) AS Missing_Quantity,
4     SUM(CASE WHEN Profit IS NULL THEN 1 ELSE 0 END) AS Missing_Profit
5 FROM superstore_sales;
6
7
```

Explanation

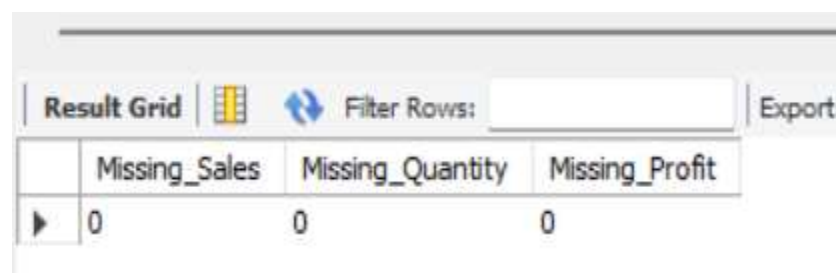
CASE WHEN

Checks whether a column contains NULL values.

SUM()

Counts the total number of missing values in each column.

OUTPUT:-



	Missing_Sales	Missing_Quantity	Missing_Profit
▶	0	0	0

Observation

If all values are 0, then the dataset does not contain missing values in these important columns.

3. Check for Duplicate Records

Duplicate records may distort analytical results and should be identified before performing business analysis.



```
1 • SELECT `Order ID`,
2       COUNT(*) AS Duplicate_Count
3 FROM superstore_sales
4 GROUP BY `Order ID`
5 HAVING COUNT(*) > 1;
6
```

Explanation

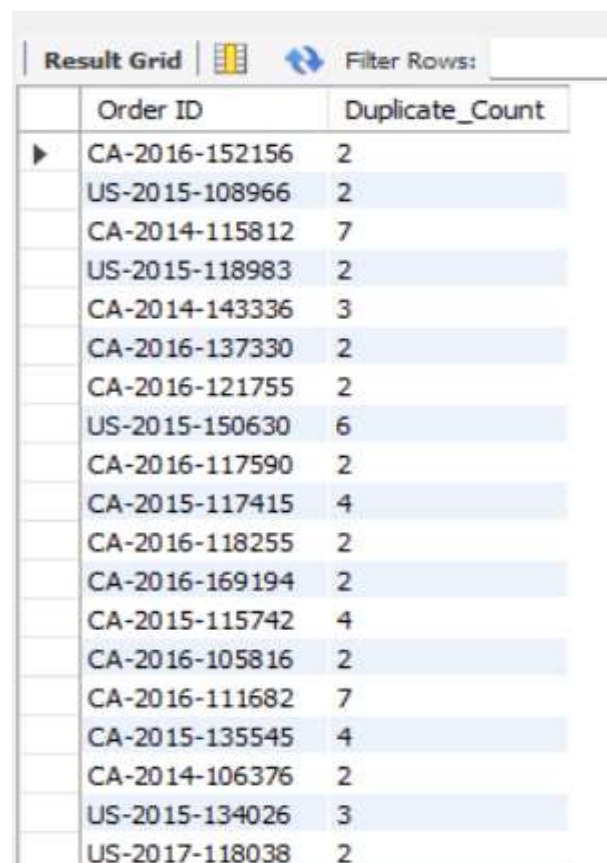
GROUP BY

Groups records using Order ID.

HAVING COUNT(*) > 1

Returns only records that appear more than once.

OUTPUT:-



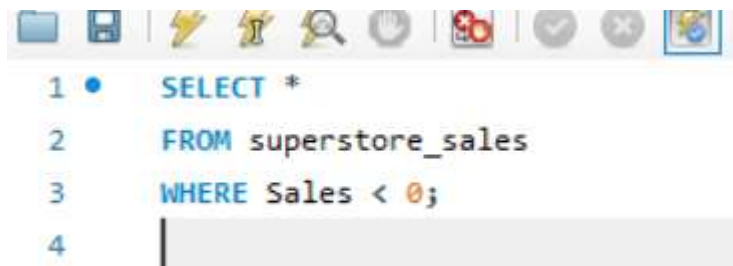
	Order ID	Duplicate_Count
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	6
	CA-2016-117590	2
	CA-2015-117415	4
	CA-2016-118255	2
	CA-2016-169194	2
	CA-2015-115742	4
	CA-2016-105816	2
	CA-2016-111682	7
	CA-2015-135545	4
	CA-2014-106376	2
	US-2015-134026	3
	US-2017-118038	2

Observation

Duplicate records were identified and reviewed to ensure data consistency.

4. Check for Negative Sales Values

Sales values should normally be positive. Negative values may indicate returns, refunds, or data entry errors.



Explanation

This query retrieves all records where the sales amount is less than zero.

OUTPUT:-

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
--------	----------	------------	-----------	-----------	-------------	---------------	---------	---------	------	-------	-------------	--------	------------	----------	--------------	--------------

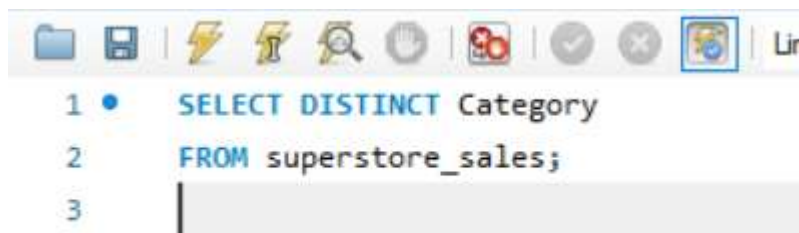
Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
-----------	-------------	---------------	---------	---------	------	-------	-------------	--------	------------	----------	--------------	--------------	-------	----------	----------	--------

Observation

Negative sales records, if present, should be reviewed to determine whether they represent returns or data issues.

5. Validate Category Values

To ensure that category values are consistent and valid.

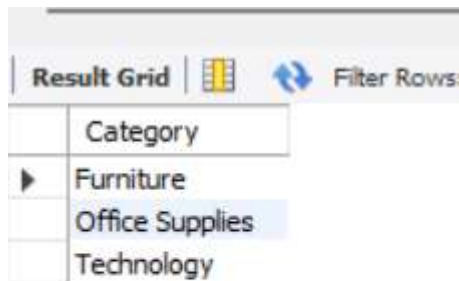


```
1 • SELECT DISTINCT Category
2 FROM superstore_sales;
3
```

Explanation

Returns all unique category values present in the dataset.

OUTPUT:-



	Category
▶	Furniture
	Office Supplies
	Technology

Observation

The dataset contains valid and consistent category values without unexpected entries.

Result

The dataset was successfully validated using row counts and data quality checks.

The following validations were performed:

1. Verified the total number of records.
2. Checked for missing values.
3. Identified duplicate records.
4. Reviewed negative sales values.

5. Validated category consistency.

Conclusion

Data validation confirmed that the Superstore dataset is suitable for analysis. The dataset was checked for completeness, consistency, and accuracy. These quality checks increase confidence in the analytical results and business insights generated in previous steps.